Assignment 6

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if you idea is (mostly) correct and you haven showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) Let the binary linear code $C \subseteq \mathbb{F}_2^5$ have parity-check matrix

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

1. Compute the syndrome of each single-bit error vector

$$e_1 = (1,0,0,0,0), \quad e_2 = (0,1,0,0,0), \quad \dots, \quad e_5 = (0,0,0,0,1).$$

2. Suppose the received word is

$$r = (1,0,1,1,1).$$

   Compute its syndrome and, assuming that at most one error occurred, decode r.

Problem 2 (25 marks) Let the binary linear code C have generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

1. List all codewords of C.
2. Determine the parameters $[n, k, d]$ of C.
3. Find a parity-check matrix H for C.
4. Suppose the received word is

$$r = (1,1,0,0,1).$$

   Compute its syndrome and decode it using the coset leader of smallest weight.

Problem 3 (25 marks) Over $\mathbb{F}_7$, consider the Reed–Solomon code of length $n = 6$ and dimension $k = 3$ with evaluation points

$$0, 1, 2, 3, 4, 5.$$

Suppose a codeword coming from a polynomial $f(x)$ with $\deg f < 3$ is transmitted, and the received word is

$$r = (2,6,5,6,2,6).$$

1. Assuming that at most one symbol error occurred, recover the transmitted polynomial $f(x)$.
2. Determine the error location and the error value.
3. Verify that the Hamming distance between the received word and the decoded codeword does not exceed the error-correcting capability of the code.

Problem 4 (25 marks) Over $\mathbb{F}_7$, consider the Reed–Solomon code with evaluation points

$$\alpha_1 = 0,\ \alpha_2 = 1,\ \alpha_3 = 2,\ \alpha_4 = 3,\ \alpha_5 = 4,$$

and message polynomials of degree < 2. Assume that at most one symbol error occurred, and the received word is

$$y = (1, 3, 0, 0, 2).$$

Let

$$E(x) = e_1 x + e_0, \qquad N(x) = a_0 + a_1 x + a_2 x^2.$$

1. Using the Berlekamp–Welch condition
$$E(\alpha_i) y_i = N(\alpha_i), \qquad i = 1, \ldots, 5,$$
write down the linear system in the unknowns e_1, e_0, a_0, a_1, a_2.
2. Find all nonzero solutions $(E(x), N(x))$.
3. Recover the transmitted polynomial
$$f(x) = \frac{N(x)}{E(x)}.$$
4. Determine the error location and the error value.